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	Complex DNA double-strand breaks along the dense ionizing track induced by HZE particles are difficult to repair and, in some instances, irreparable. These unrepaired and/or misrepaired DNA lesions may contribute to the observed higher relative biological effect (RBE) for cell killing, chromosomal aberrations, mutagenesis, and carcinogenesis in cells irradiated with HZE particle. While the types of DNA lesions induced by high LET radiation have been well studied in vitro, it is still unclear why HZE particles induced DNA lesions are difficult to repair or why the current pathways of DSB repair cannot handle these lesions properly. The goals of this proposal are two fold: 1) to develop new approaches to determine the types of DNA lesions formed and their spatial relationship along the dense ionizing track induced by different HZE particles in human cells, while trying to discern the cellular capability to repair these DNA lesions in situ; and 2) to identify the molecular pathways (nonhomologous end joining and homologous recombination) or factors		

(Artemis and/or WRN) that may be able to manage HZE particle-induced DNA lesions including DNA DSBs in human cells. In order to achieve these goals, we propose: 1) to verify that multiple types of DNA lesions formed in close vicinity along the dense ionizing tracks induced by HZE particles in situ; 2) to test the hypothesis that Artemis (a specific exo-/endonuclease) may be involved in the repair of complex DNA DSBs induced by HZE particle through the nonhomologous end joining pathway; 3) to test the hypothesis that WRN, a human multifunctional protein with exonuclease and helicase activities, may be involved in managing the repair of HZE induced DNA DSBs; and 4) to determine the role of HR in the repair of complex DSBs induced by HZE particles. Determining the biochemical nature of the DNA damages induced by HZE particles and identifying the factors that are required for the repairing of these DNA lesions will provide tools to assess individual radiation susceptibility and to validate risk assessment for human exposure to HZE particles. Ultimately, our research would lead to developing counter-measures to enhance astronauts' ability to sustain HZE particles and radiation exposures in space.
Description:

Task Description:

Overall Objective

DNA repair capability is one of the key indicators reflecting the biological consequences of cellular exposure to genotoxic insults. Determining the biochemical nature of the DNA damages induced by HZE particles and identifying the factors that are required for the repairing of these DNA lesions will provide tools to assess individual radiation susceptibility and also to validate risk assessment for human exposure to HZE particles. The goals of this proposal are two fold: 1) We will develop new approaches to determine the types and spatial relationship among different DNA lesions formed along the dense ionizing tracks induced by HZE particles in human cells and to dissect the cellular capability to repair these DNA lesions in situ; and 2) we will identify molecular pathways or factors that may be able to help the human cell respond efficiently to HZE particle-induced DNA lesions. For this purpose, we will test the hypothesis that specific nucleases, Artemis and/or Werner Syndrome protein (WRN), contribute specifically to the repair of complex DNA lesions induced by HZE particles. We will also determine the role of homologous recombination factors in repair of HZE particle-induced DSBs. With these new approaches on hands, we should be able to provide a semi quantitative measurement for the generation and spatial distribution of clustered DNA lesions along the dense track induced by HZE particles and determine the relative contribution of these Nonhomologous end-joining and homologous end joining factors in the repair of DNA DSBs, but also able to identify whether these repair activities play any role in managing clustered DNA lesions generated by HZE. These studies would, in turn, provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future.

Specific Objectives

A.1. To verify that multiple types of DNA lesions formed in close vicinity along the dense ionizing tracks induced by HZE particles in situ. We will use antibodies for repair enzymes, fluorescent tagged repair proteins as surrogate markers as well as direct assay system to identify various types of DNA lesions and also monitor the repair status of the DNA damage in human cells exposed to HZE particles.

1) To verify whether DNA lesions including base damage, DNA single strand breaks (SSB), and DSBs are clustered along the tracks induced by HZE particles.

2) To determine the repair of clustered DNA lesions formed along the tracks induced by HZE particles.

3) To determine the nature of the persistent or “unrepaired” DNA lesions induced by HZE and its spatial relationship to clustered DNA damages.

A.2. To test the hypothesis that Artemis, a specific exonuclease/endonuclease, which may be involved in the repair of complex DNA DSBs induced by HZE particle through nonhomologous end joining (NHEJ) pathway. Evidence suggest that Artemis protein, a specific 5’ to 3’ exonuclease/endonuclease, which play a key role in NHEJ, but specially play a role in the repair of complex DNA DSBs induced by high LET radiation. We will determine the functions of Artemis in the repair of DSBs induced by HZE particles.

1) To determine the radiosensitivity and repair of DNA DSBs and clustered DNA lesions induced by various HZE particles in human Artemis deficient cell lines.

2) To determine whether the Artemis’ exonuclease/endonuclease activity is involved in the repair of HZE induced DSBs. We will measure the radiosensitivity and repair of DSBs induced by HZE in human Artemis nuclease dead cell line.

3) To determine the role of Artemis in recognition of HZE particle induced DSBs. We will use green fluorescent protein (GFP)-tagged Artemis to measure the initial and kinetics of the Artemis localization on DNA damage sites induced by various HZE particles.

A.3. To test the hypothesis that WRN, a human multifunctional protein with both exonuclease and helicase activities, protein may involve in managing the repair of HZE induced complex subset of DNA DSBs. Our preliminary data suggests that human Werner syndrome patient cells exposed to HZE particles result in elevated unrepaired DSBs as compared to those induced by gamma-rays. However, the role of WRN in DSB repair, especially in HZE induced DSBs, is not clear. We plan to determine the role of WRN in processing complex DSBs in HZE irradiated cells.

1) To determine the radiosensitivity and repair of DNA DSBs and clustered DNA lesions induced by various HZE particles in human WRN deficient cell lines.

2) To determine the roles of exonuclease and helicase activities of WRN in repair of HZE particles induced DNA DSBs and clustered DNA lesions.

3) To determine whether WRN localization on DNA damage sites is dependent on the quality of HZE particles and also whether WRN phosphorylation is required for the repair of HZE induced DSBs.

A.4. To determine the role of homologous recombination (HR) in the repair of complex DSBs induced by HZE particles. Indirect evidence suggests that HR frequency might be increased in cells exposed to high LET radiation. However, the role of HR in the repair of HZE particle induced DSBs in human cells is completely unknown. We will use unique human knockout mutant cell lines to address this issue.

1) To determine whether eliminating both HR and NHEJ pathways would elevate unrepaired DNA DSBs and enhance

	cellular sensitivity to HZE particle irradiation. We have constructed a set of isogenic knockout human cell lines that are deficient in HR (Rad54-/-), NHEJ (ligase IV-/-) and HR/NHEJ (Rad54-/- and Ligase IV-/-), and will determine the radiation sensitivity and DSB repair of these cell lines in response to HZE particles. 2) To determine whether the function of WRN in response to HZE particle is involved in HR or NHEJ of the DNA DSB repair pathway. In order to answer this question, we plan to measure whether there is a synergistic radiobiological effect in WRN-/-/Rad54-/- or WRN-/-/ligase IV-/- double knockout human cell line as compared to individual knock out cell lines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research would lead to the development counter-measurements to enhance astronaut's ability to sustain HZE particles and radiation exposures in space.
Task Progress:	1. To study induction and repair of clustered DNA damages, we irradiated HT1080 cells stably expressing green-fluorescent protein tagged XRCC1 (EGFP-XRCC1) with Fe, Si and Oxygen particles and immunostained with 53BP1 (double strand breaks) and OGG1 (base damage). Using these as surrogate markers, we noticed differential induction of clustered DNA damages by different HZE particles. Moreover, the number of cells harboring the un-repaired clustered damages was higher in Fe irradiated cells, as compared to Si and O. 2. To study the differential induction of different types of DNA damages, including DSBs, SSB and base damages, along the ion tracks traversed by HZE particles, we have established stable cells lines expressing RFP-XRCC1 and YFP-53BP1 and confirmed their recruitment to the sites of DNA lesions generated by Fe, Si particles and gamma-rays. Using this approach, we directly monitored induction of single and DSBs in single living cells. 3. To study the genomic instability caused by charged particles, we irradiated HT1080 cells with Fe particles (1Gy, 1GeV/n) and examined the chromosomal aberrations 24 hours after irradiation. When we analyzed the metaphase spreads, we noticed different types of chromosomal abnormalities in Iron irradiated cells as compared with mock-irradiated cells. 4. To determine the cell cycle delay induced by charged particles, we irradiated exponentially growing HT1080 cells with Fe particles and collected samples before (pre-IR) and 0.5, 1, 2, 4, 8, 12, 24 and 48 hours after irradiation and checked for cell cycle delay by Flow cytometer. Our results showed that irradiation of cells with Fe ions causes extended period of G2/M arrest.
Bibliography Type:	Description: (Last Updated: 08/14/2018)
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